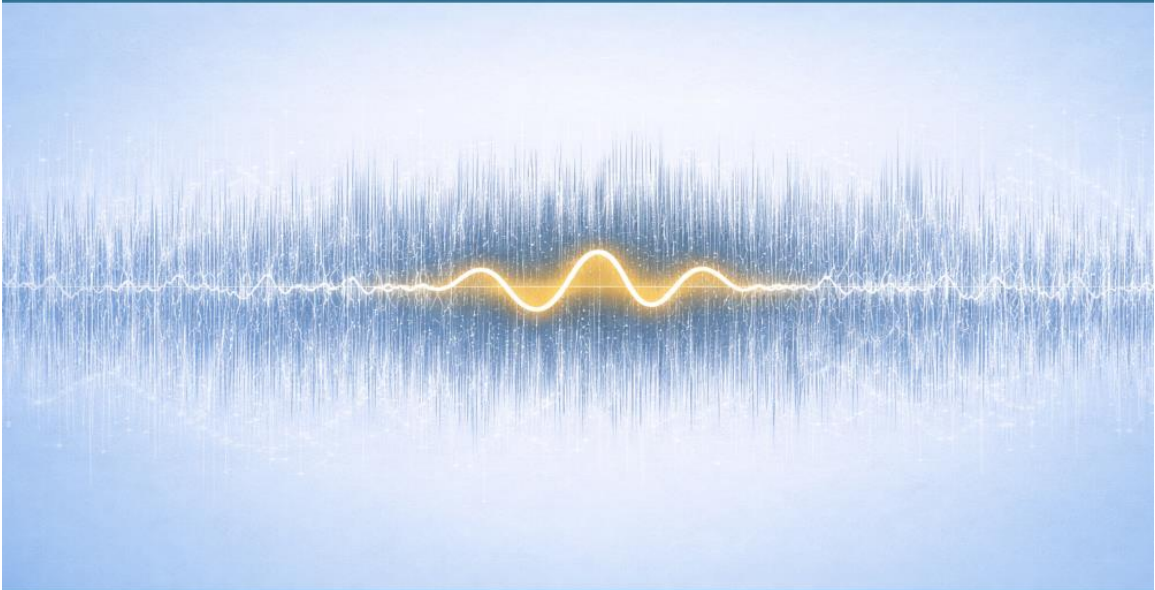


Lock-In Amplifier From Theory to Simulation



Leonardo Chieco

Lock-In Amplifier

From Theory to Simulation

The lock-in amplifier is a fundamental instrument in electronics and laboratory instrumentation, designed to detect weak signals buried in high levels of noise. By exploiting the principle of synchronous multiplication combined with selective filtering, it allows the extraction of a signal at a known frequency, even when its amplitude is much smaller than the surrounding noise. This technique is widely used in fields such as spectroscopy, photoacoustics, and the analysis of modulated signals, where measurement accuracy and reliability are essential.

After a theoretical introduction, a simulation of a lock-in amplifier will be presented, enabling direct observation of its operating principles and providing a clear understanding of circuit behavior before moving on to practical implementation.

Enjoy the reading!

The Author

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1. Fundamentals and Purpose of the Lock-In Amplifier

The lock-in amplifier, or synchronous detector, is an extraordinarily effective instrument for measuring weak signals buried in noise, as it exploits the frequency and phase coherence between the signal of interest and a known reference. In practice, instead of amplifying everything indiscriminately, the lock-in “selects” only the component of the signal that oscillates at the correct frequency and maintains the expected phase, rejecting everything else, including random noise.

At the core of the lock-in lies the synchronous multiplier, also known as the Phase Sensitive Detector (PSD).

Before proceeding, let us recall the trigonometric identities for sine and cosine multiplication:

$$\sin(\alpha) \cdot \cos(\beta) = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

$$\sin(\alpha) \cdot \sin(\beta) = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

Let us denote the useful signal as $s(t)$ and the reference as $r(t)$:

$$s(t) = A \sin(\omega t + \phi) , \quad r(t) = \sin(\omega t)$$

By multiplying the signal and the reference, we obtain:

$$s(t) \cdot r(t) = A \sin(\omega t + \phi) \cdot \sin(\omega t) = \frac{A}{2} \cos(\phi) - \frac{A}{2} \cos(2\omega t + \phi)$$

From this expression, two components emerge: one at double frequency (2ω) and one DC component, independent of time. By applying a low-pass filter after the multiplier, the 2ω term is removed, leaving a stable DC output proportional to the amplitude of the useful signal.

Now consider a signal $s(t)$ corrupted by noise $n(t)$ of arbitrary amplitude:

$$s(t) = A \sin(\omega t + \phi) + n(t)$$

$$\begin{aligned}
 s(t) \cdot r(t) &= (A \sin(\omega t + \phi) + n(t)) \cdot \sin(\omega t) \\
 &= \frac{A}{2} \cos(\phi) - \frac{A}{2} \cos(2\omega t + \phi) + n(t) \cdot \sin(\omega t)
 \end{aligned}$$

After the low-pass filter, the DC term $\frac{A}{2} \cos(\phi)$ remains, effectively free of noise.

The conceptual block diagram of a lock-in amplifier is shown in the following figure.

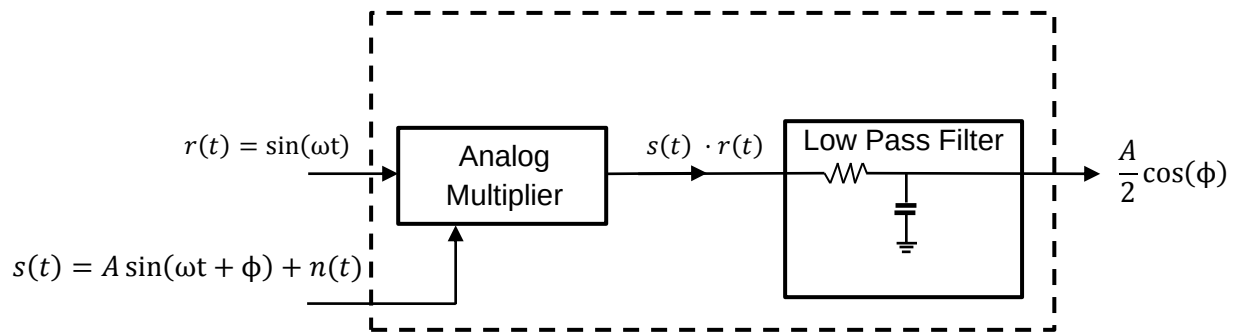


Figure 1: Conceptual diagram of a lock-in amplifier

The lock-in exploits the fact that noise is random and uncorrelated with the reference. When the signal is multiplied by the reference, the non-coherent components average to zero, while the coherent signal emerges as a stable DC value. This makes it possible to detect signals that are orders of magnitude smaller than the background noise.

An intuitive analogy is that of a perfectly tuned radio: only the desired frequency is amplified, while everything else is rejected. In this sense, the lock-in is not simply an amplifier, but a highly selective filter based on frequency and phase, capable of providing precise information about the amplitude and phase of signals that would otherwise be undetectable with conventional instruments.

In practical terms, this technique is widely used in spectroscopy, optical detection, impedance measurements such as EIS, and in any application where the signal of interest is deeply buried in noise.

2. Lock-In with Quadrature Signals

There are situations where it is necessary to measure the phase shift ϕ between the signals $s(t)$ and $r(t)$, or where the amplitude measurement must be independent of phase.

This problem is solved by implementing a lock-in amplifier with two quadrature channels (sine and cosine). One channel multiplies the signal by an “in-phase” reference $\sin(\omega t)$, while the other uses a “quadrature” reference (shifted by 90°) $\cos(\omega t)$. The outputs, commonly referred to as X (in-phase) and Y (quadrature), allow both the amplitude and phase of the signal to be determined.

Let us see how.

Assume:

$$s(t) = A \sin(\omega t + \phi) \text{ input signal}$$

$$r_I(t) = \sin(\omega t) \text{ in-phase reference}$$

$$r_Q(t) = \cos(\omega t) \text{ quadrature reference}$$

For the in-phase channel:

$$\begin{aligned} X(t) &= s(t) \cdot r_I(t) \\ &= A \sin(\omega t + \phi) \cdot \sin(\omega t) \\ &= \frac{A}{2} [\cos((\omega t + \phi) - \omega t) - \cos((\omega t + \phi) + \omega t)] \\ &= \frac{A}{2} [\cos(\phi) - \cos(2\omega t + \phi)] \end{aligned}$$

For the quadrature channel:

$$Y(t) = s(t) \cdot r_Q(t) = A \sin(\omega t + \phi) \cdot \cos(\omega t) = \frac{A}{2} [\sin(\phi) + \sin(2\omega t + \phi)]$$

In both cases, by extracting only the DC component through a low-pass filter, we obtain:

$$X_{DC} = \frac{A}{2} \cos(\phi)$$

$$Y_{DC} = \frac{A}{2} \sin(\phi)$$

Observing that:

$$X_{DC}^2 + Y_{DC}^2 = \frac{A^2}{4} (\sin^2(\phi) + \cos^2(\phi)) = \frac{A^2}{4}$$

since $\sin^2(\phi) + \cos^2(\phi) = 1$, we can derive the amplitude:

$$A = 2 \sqrt{X_{DC}^2 + Y_{DC}^2}$$

For the phase ϕ , we note that:

$$\frac{Y_{DC}}{X_{DC}} = \frac{\sin(\phi)}{\cos(\phi)} = \tan(\phi) \rightarrow \phi = \arctan \frac{Y_{DC}}{X_{DC}}$$

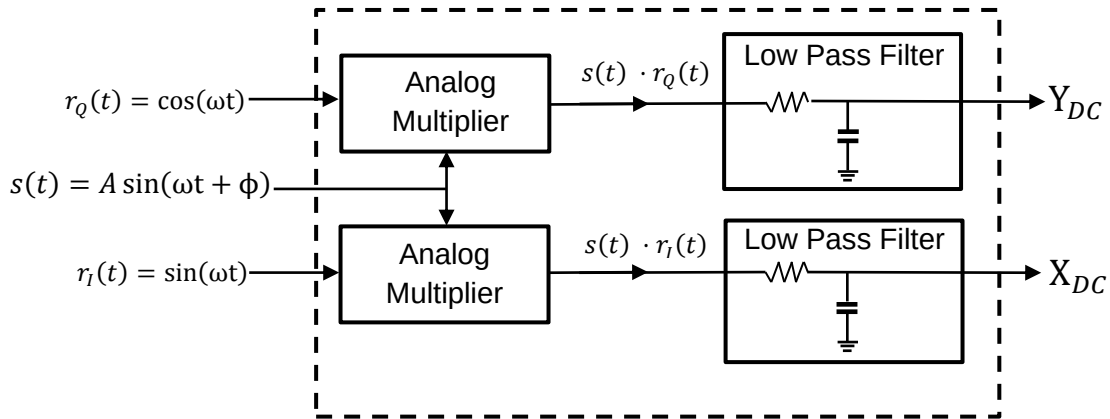


Figure 2: Lock-in with quadrature signals

In modern systems, the voltages Y_{DC} and X_{DC} are acquired through an ADC, and the amplitude and phase are computed digitally by a microcontroller.

3. Lock-In Simulation

Before going into the details of the simulation, it is important to understand why the lock-in amplifier is such a relevant tool in practice. This technique is used whenever a very weak signal must be measured in the presence of noise. In all these cases, the lock-in allows the signal of interest to be “locked” to a known frequency and selectively extracted, dramatically improving the signal-to-noise ratio.

For this reason, simulation plays a fundamental role, as it allows us to observe in a controlled way what happens inside the circuit, isolating individual phenomena (multiplication, frequency translation, filtering) that in reality occur simultaneously. Without simulation, many of these effects remain difficult to grasp intuitively.

In the LTspice simulation, an analog multiplier AD633 was used to implement a simple lock-in amplifier. This integrated circuit, produced by Analog Devices, is designed to perform analog multiplication between signals. It is a four-quadrant multiplier, meaning it can multiply both positive and negative signals. Its characteristic equation is:

$$W = \frac{(X1 - X2)(Y1 - Y2)}{10} + Z$$

where:

- X1, X2, e Y1, Y2 are the differential inputs,
- Z is an offset input (useful for summing a signal),
- W is the output.

Its main features include a typical supply of ± 15 V (it also works at lower voltages), a bandwidth of about 1 MHz, an internally scaled gain (factor 1/10), and the fact that it does not require external components to operate.

For further details, refer to the datasheet.

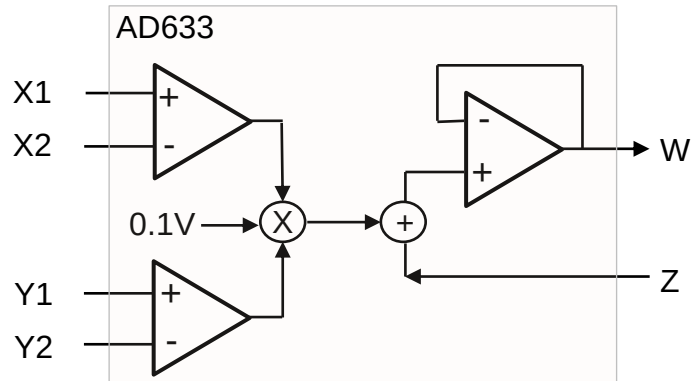


Figure 3: Internal schematic of the AD633 integrated circuit

Before analyzing the lock-in circuit, it is useful to frame a typical measurement problem. Suppose we excite a Wheatstone bridge (e.g., a pressure sensor) with a sinusoidal signal at 32 kHz, while the physical quantity to be measured varies slowly over time, for example with a sinusoidal behavior at 10 Hz. Under these conditions, the signal of interest appears as a high-frequency carrier (32 kHz) amplitude-modulated by a low-frequency signal (10 Hz), as shown in the following figure.

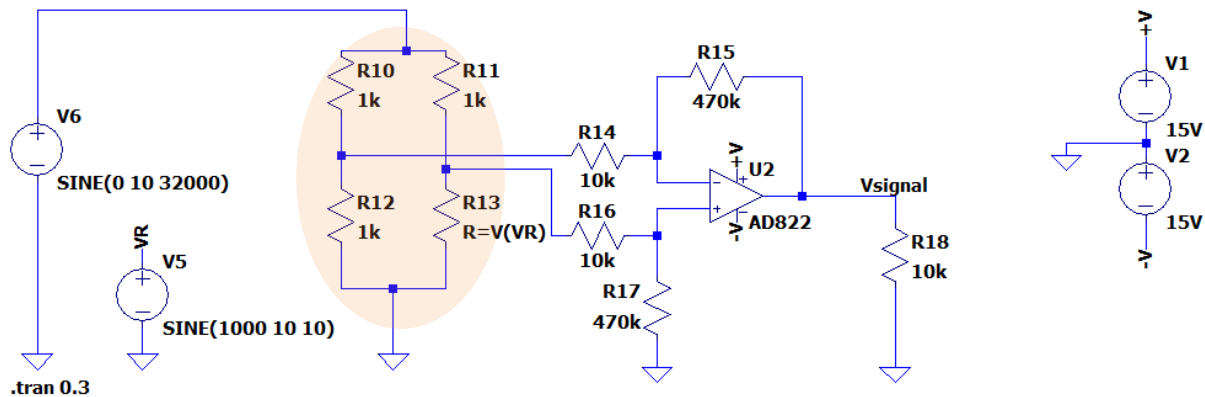


Figure 4: Wheatstone bridge (highlighted) driven by a sinusoidal voltage

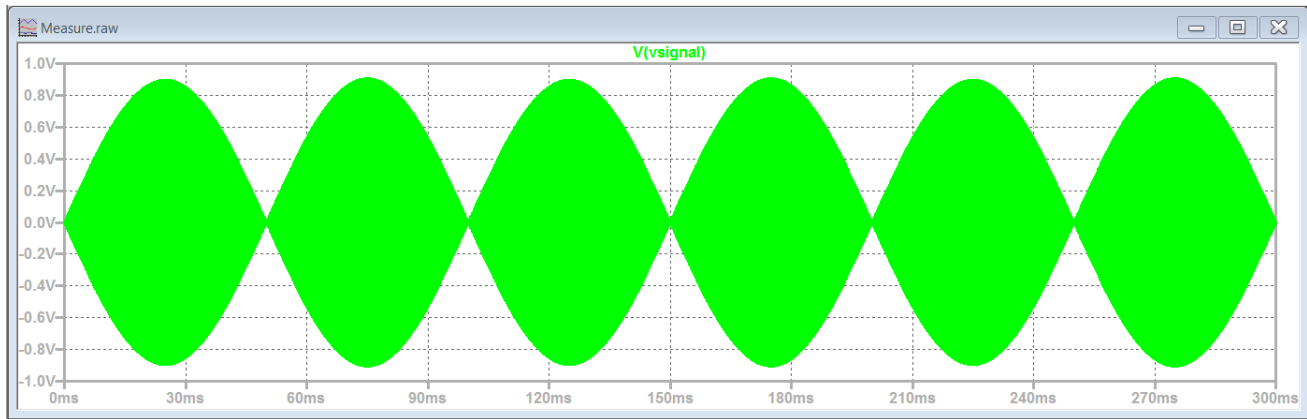


Figure 5: Output waveform

In the simulation, the value of resistor R4 is made variable to emulate the change in the physical quantity, by linking it to a sinusoidal voltage source with a mean value of 1000 V and an amplitude of 10 V. As a result, the resistance oscillates around its nominal value, ranging between 990 Ω and 1010 Ω ($R4 = 1000 \pm 10 \Omega$).

A differential amplifier (U2) amplifies the weak signal provided by the bridge and brings it to approximately 1.8 Vpp.

To keep the simulation lightweight without compromising its physical meaning, the previously shown circuit can be replaced by an equivalent voltage source obtained as the product of two signals: a pure 32 kHz sine wave and a 10 Hz sine wave. The resulting signal is therefore a 32 kHz carrier amplitude-modulated at 10 Hz, fully equivalent, from a functional standpoint, to the waveform observed in the original simulation. A white noise signal with an amplitude of 20 Vpp is then added to reproduce a realistic and highly noisy measurement condition.

Two signals are then applied to the AD633 inputs X1 and Y1: on one side, the pure 32 kHz sine wave used as the reference (and coinciding with the bridge excitation), and on the other side, a signal at the same frequency but amplitude-modulated, representing the useful information, i.e., the Wheatstone bridge output immersed in noise.

Inputs X2 and Y2 are connected to ground since single-ended signals are used instead of differential ones. The offset input Z is also grounded, as it is not required in this case.

The output of the multiplier is filtered using a second-order Sallen-Key low-pass filter with a cutoff frequency of approximately 12 Hz, designed to remove the high-frequency components resulting from the multiplication. The DC component is proportional to the signal amplitude; since this amplitude is modulated sinusoidally at 10 Hz, the expected output is a sinusoidal signal at the same frequency of 10 Hz.

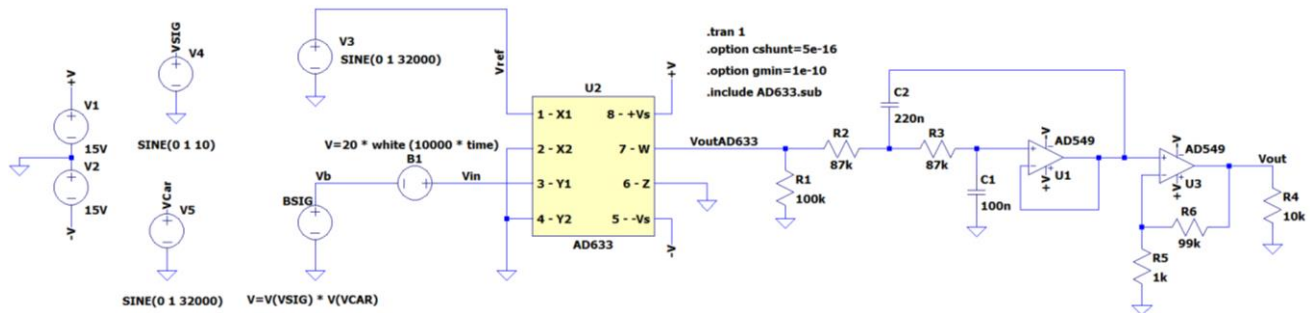


Figure 6: Lock-in schematic used for the simulation

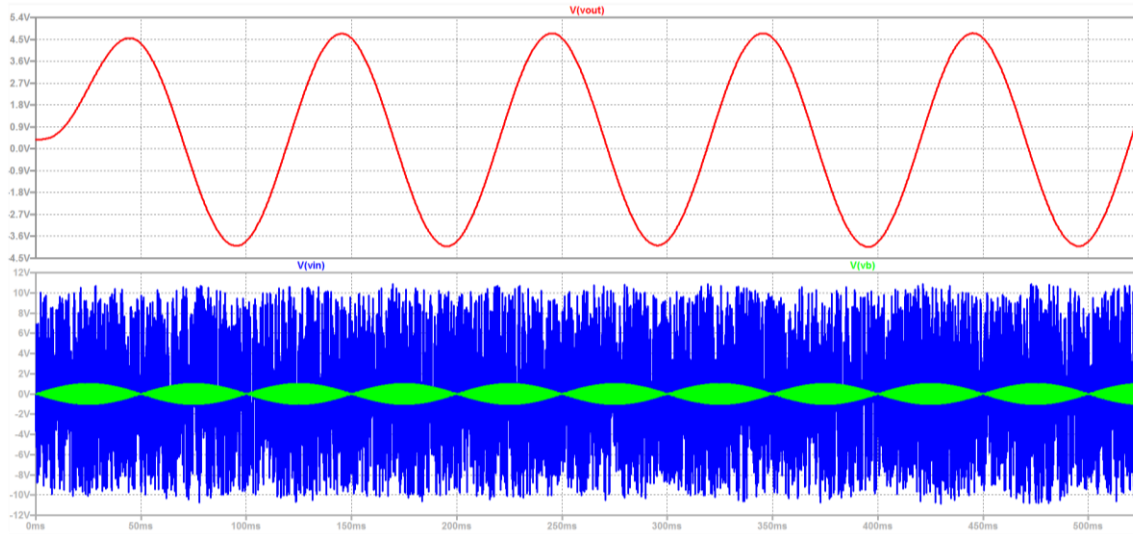


Figure 7: Simulation waveforms

The simulation results clearly show that the 10 Hz modulating signal is perfectly reconstructed and isolated despite the strong noise, while all other components are effectively attenuated by the filter. This example provides a practical demonstration of the effectiveness of the lock-in amplifier in extracting a weak signal from a high-frequency carrier and surrounding noise, confirming the theoretical principles of synchronous multiplication and selective filtering.

4. Conclusions

The simulation clearly demonstrates the effectiveness of the lock-in amplifier in extracting weak signals embedded in high levels of noise. By using the AD633 analog multiplier together with a Sallen-Key low-pass filter, the low-frequency modulating signal is accurately reconstructed, while high-frequency components and surrounding noise are effectively attenuated. This practical example confirms the theoretical principles of synchronous multiplication and selective filtering, showing how the lock-in can transform complex and noisy measurements into clear and reliable data.